

Intersection between surfaces using Computer Extended Descriptive Geometry (CeDG): application to the focal illumination of a sphere

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Abstract Computer Extended Descriptive Geometry (CeDG) is a new approach to computer modelling of 3D geometric systems that tries to overcome several limitations of current CAD systems. A preliminary version of CeDG for GeoGebra has demonstrated advantages in sheet metal and mechanisms field. This paper develops the theoretical basis of the Locus – based Surfaces’ Intersection Method (LSIM) of CeDG and compares it against the standard Descriptive Geometry technique, through the calculation of the illumination of a sphere by a focal light beam. Results showed that, in opposition to standard Descriptive Geometry technique, the LSIM CeDG model combines less complexity (geometric procedure requires only one iteration) with the capability to be extended to other parameters’ values and projections, keeping the compliance with geometrical requirements. Accuracy metrics have demonstrated that LSIM can generate the exact (true) surface intersection curve through its projections, thanks to the geometric integrity of the underlying dynamic geometry software (GeoGebra).

Keywords: Descriptive Geometry, CAD, Computer Parametric Graphic Modelling, Dynamic Geometry Software, CeDG.

1 Introduction

Descriptive Geometry (DG) and its derived representation systems, such as dihedral system, have defined the fundamentals of graphical representation in engineering since the second half of the 19th century, establishing an important scientific-technical corpus [1]. This one includes many texts that continue to provide a solid basis for university teaching [2-7], even though their use in the professional field has been practically replaced by computer aided design software (CAD) [8].

The geometry of CAD model is mainly represented by fit curves and surfaces, such as B-splines, which provide high control and accuracy [9,10].

Computer Extended Descriptive Geometry (CeDG) is a new approach to computer modelling of 3D geometric systems [11], which tries to address the limitations of current CAD systems pointed out by some authors [11-13]. Briefly, CAD tools allow the construction of virtual prototypes of 3D systems that can be manipulated in space and easily projected according to the chosen representation system. However, they do not facilitate the creation of the model when it depends on some implicit parameter, in addition to presenting shortcomings in the calculation of flat patterns of sheet metal surfaces.

The CeDG approach combines the ability of DG to solve spatial geometric problems with the ability of dynamic geometry in the process of building geometric-algebraic models [14-16]. Unlike CAD systems, CeDG parametric models preserve the integrity of curves and surfaces. A preliminary version of the CeDG approach implemented on the dynamic geometry software GeoGebra™ [17] has demonstrated its ability to overcome some of the above limitations [11,18].

The objective of this study is to present and analyze the method used in CeDG models to obtain the intersection curve between two surfaces. The usual procedure in DG consists of defining it by means of its projections, which are calculated in turn by interpolation on a set of points belonging to them. The points are obtained by means of DG techniques, applied iteratively. Accordingly, the accuracy and complexity of the resulting curve is proportional to the number of points obtained [7]. The CeDG approach uses an extension of the DG procedures, which allows the generation of a specific locus function for each projection of the intersection curve sought. The locus function is automatically determined from the sequence of geometric-algebraic instructions associated with the calculation of a single generic point of the intersection [11]. This paper develops the theory that defines the foundation of the Locus – based Surfaces' Intersection Method (LSIM) and compares this one with respect to the standard DG procedures through a case study.

2 Methods

The study is developed according to the following workflow:

- i. The theory of LSIM is developed using the 2D locus function of the dynamic geometry software as algebraic support. Several examples are succinctly presented to describe two variants of this novel method.
- ii. A case study has been defined to evaluate the goodness of LSIM. This is a sphere illuminated by a focal (spot) light beam. The problem is solved using both standard descriptive geometry procedures and the LSIM – based CeDG approach.

- iii. Solutions to the illuminated surface reached in the previous point are compared to give the methodological advantages and the improvement in accuracy of CeDG against the standard descriptive geometry procedures.

The novel LSIM requires the development of a theory that is succinctly presented in the following Section. The examples selected to support the theoretical description include a cone – cylinder intersection and the focal illumination of the sphere that is subsequently executed as case study. The case study is defined with detail in this Section with the aim of facilitating the definition of metrics for the comparison stage.

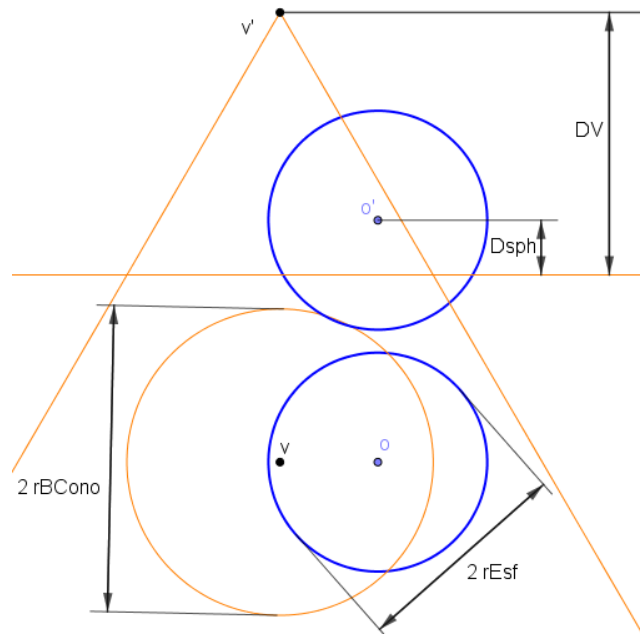


Fig. 1. Sphere to be illuminated from a focal light beam with spot in $v-v'$ and cone surface defined by the spot as vertex, vertical axis and circular directrix.

The spatial system that defines the case study appears in Fig. 1 through the projections. The focal spot that illuminates the sphere is above the horizontal plane that contains the circular directrix of the light cone (distance DV), which in turn has a diameter of $2 \cdot r_{BCono}$. The diameter of the sphere is $2 \cdot r_{Esf}$ and its center $o-o'$ is above the previous horizontal plane at the distance $Dsph$. These parameters control completely the focal illumination of the sphere through a conical light beam with solid angle equal to Ω .

Table 1. Main parameters of the sphere illuminated from a focal light beam.

<i>Parameter</i>	<i>Definition</i>	<i>Default value (mm)</i>
rBCono	Radius of light cone section at the reference horizontal plane	70
rEsf	Radius of the illuminated sphere	50
DV	Distance of focal spot to the reference horizontal plane	120
Dsph	Distance of sphere center to reference horizontal plane	20

Those main parameters are defined in Table 1, together with the default value that was used during the model building process. The value of the solid angle of the conical light beam may be written as follows:

$$\Omega = \frac{DV}{\sqrt{DV^2 + rBCono^2}}, \quad (1)$$

from which the default value of Ω is 0.27π sr.

The illumination of the sphere is computed through the warped curve in the sphere that encloses the lightened zone.

Concerning the metrics, we use the next qualitative properties to compare the CeDG methodology against that of the standard descriptive geometry procedures:

1. Complexity of the method and number of iterations required to get the proper solution.
2. Additional procedures required to guarantee that the solution is compliant with geometrical and physical requirements.
3. Capability of the methodology to extend the solution to other parameters' values and projections.

The sought warped curve that defines the boundary of the lightened zone in the sphere is composed by a piece of the cone – sphere intersection and an arc of the separatrix circle associated with the light spot. We evaluate the accuracy of the cone – sphere intersection, taken advantage that the projection of this curve in the vertical plane of Fig. 1 is an arc of parabola. The reason is that both surfaces are revolution quadrics with axes defining a plane parallel to vertical plane [6]. The metrics that quantify the comparison of accuracies are extracted as follows:

1. The conic associated to the true (exact) cone – sphere intersection projection is computed using five true points of this conic, in the dynamic geometry software (GeoGebra).

2. We calculate the dispersion of the conics identified by GeoGebra when their definition points are moved along the cone – sphere intersection projection, from the true point towards its extreme points, defined as equidistant points from their adjacent true points. The procedure is clarified in Fig. 2, which shows a true point and their two extreme points M_1 and M_2 . The true points and projection curve have been extracted and enlarged from Fig. 9.

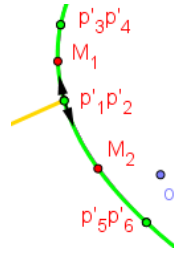


Fig. 2. True point ($p'1p'2$) in the projection of cone – sphere intersection and points M_1 and M_2 , which are equidistant to the adjacent true points ($p'3p'4$, $p'5p'6$).

3. The horizontal distance between any point in the calculated projection curve and the true projection curve (absolute error curve), $Err(y)$, is calculated as a function of the vertical distance, y .
4. The influence of dimensional parameters of the system in $Err(y)$ is obtained and discussed.

3 Locus - based surface intersection theory

The intersection between two surfaces produces a curve in space that in general will not be flat. The calculation of these curves is a ubiquitous problem in science and engineering, including the 3D definition of any industrial part or system, the spatial analysis of biomechanical systems, or the study of a focal lighting system, to name a few examples.

The general technique for calculating the intersection curve between surfaces in descriptive geometry is based on the use of a set of auxiliary surfaces, defined in such a way that the encounter between any one of them and the two data surfaces produces a pair of simple curves that will be cut by belonging to the same auxiliary surface, to provide points of the intersection curve sought. This procedure must be repeated until enough points are obtained to define the curve with the required accuracy.

The details concerning the typology of auxiliary surfaces to be used depending on the datum surfaces are part of the large body of existing knowledge in descriptive geometry. One feature to note in this general technique is that the points that meet each auxiliary surface are exact (true). This property provides some control over the accuracy of the resulting 3D curve.

This procedure on paper is iterative and time-consuming and requires obtaining notable points of the curve, such as those belonging to contour lines and auxiliary boundary surfaces, which ensure that it meets a set of minimum quality criteria. As the contour lines depend on the direction and type of projection, the points of contact with the desired curve also depend on the direction and type of projection, which is an additional shortcoming of the technique.

The above limitations are inherent to the manual technique of performing the procedures, but not to the descriptive geometry itself. Thus, the general technique of calculating the intersection curve can be formulated as follows. Calling σ the plane curve resulting from the projection of the intersection curve C onto a plane of interest, σ will be defined by the set of all its points p_σ . On the other hand, if the auxiliary surfaces that give the points p_σ of σ are expressed as $S(\omega)$, where the parameter ω in a set Ω identifies each of the possible surfaces, then the curve σ can be expressed as the locus of the points $p_\sigma(\omega)$ for all ω . That is:

$$\mathcal{L}(\omega) \equiv \sigma = \text{locus}(p_\sigma(\omega), \omega \in \Omega), \quad (2)$$

where $\mathcal{L}(\omega)$ is a function that represents the curve parametrically, providing points in flat space (plane on which C is projected) and taking $\omega \in \Omega$ as a parameter. The function $\mathcal{L}$ exists and will be continuous under certain usual continuity assumptions on $S(\omega)$ and on data surfaces. Under these conditions, CeDG allows to construct the functions $\mathcal{L}_i$ associated with the projections of the intersection curve C onto the planes of interest denoted by the index i (σ_i), using the Geogebra locus command.

This command is mainly employed in two forms: with a reference point, locus (p_σ, p_{ref}) and with a parameter, locus (p_σ, t). In the first form, the set Ω is defined by a reference locus with points p_{ref} , while the second form sets Ω as an interval in $\mathbb{R}$ defining the values of t . To clarify the constructive procedure in each case, two examples are discussed below.

3.1 Ω defined as a reference locus

The first example uses the cylinder-cone bite of Fig 3. The intersection between these surfaces is a warped curve that can be obtained on paper by the general technique of intersection between surfaces of descriptive geometry, but the process is very sensitive to drawing errors due to the nature of the curve. Fig. 3 shows three projections of such a curve, obtained using the LSIM – based CeDG model.

Auxiliary surfaces used for this system are planes defined by the vertex of the cone (V) and one of the generatrices of the cylinder. Since the cylinder is perpendicular to the profile plane, its projection in this plane coincides with its circular directrix. Fig 3 shows a generic auxiliary plane perpendicular to the profile plane and defined by the line passing through the points v'' and $3''4''$ in the profile. This auxiliary plane allows obtaining two points of the intersection curve sought, defined by 3 and 4 in the horizontal projection, and $3'$ and $4'$ in the vertical one.

To build the LSIM – based CeDG model of this system, we start by calculating the points 3 and 4 in 3D starting from the auxiliary plane mentioned above, defined in terms of a free point on the reference locus. The free point is $3''4''$ (p_{ref}) and the reference locus is the arc of circle inside the cone between the boundary points $1''$ and $2''$ in the profile, as shown in Fig 3. This point can be created graphically by clicking on the arc after selecting the point tool (**Point**(Arc) command). Once created, it can be dragged with the mouse along the arc. Once the chosen plane is placed in a comfortable situation to work (preferably an intermediate one), the general intersection technique is applied, which will provide the points 3 and 4 belonging to the intersection curve sought, symmetrically placed on the left and right side of the plan and elevation views.

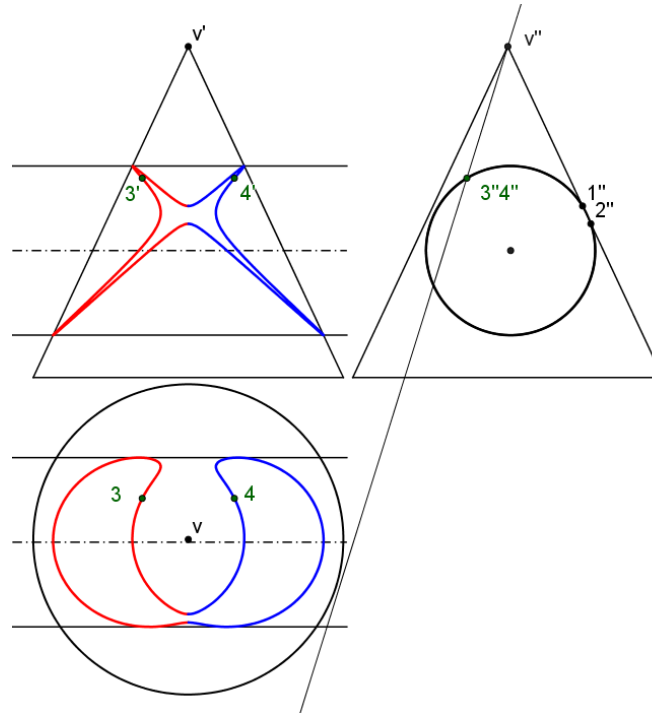


Fig 3. LSIM - based CeDG model of cylinder-cone bite.

According to the previous analysis, the arc of circle is the Ω locus that defines the projections of the intersecting curve according to equation (2). Thus, the horizontal projection of the curve, σ_1 , will be:

$$\mathcal{L}_1(p_{\text{ref}}) = \sigma_1 = \begin{cases} \text{locus}(3, p_{\text{ref}}), & \text{left leaf} \\ \text{locus}(4, p_{\text{ref}}), & \text{right leaf} \end{cases} \quad (3)$$

where 3 and 4 denote the horizontal projections of the points obtained through the auxiliary plane. In the same way the vertical projection will be:

$$\mathcal{L}_2(p_{\text{ref}}) = \sigma_2 = \begin{cases} \text{locus}(3', p_{\text{ref}}), & \text{left leaf} \\ \text{locus}(4', p_{\text{ref}}), & \text{right leaf} \end{cases} \quad (4)$$

where 3' and 4' are the vertical projections of the points obtained through the auxiliary plane.

Equations (3) and (4) define the cylinder-cone intersection curve by means of algebraic entities created in the process of determining the pair of points associated with the generic auxiliary plane, thus defining an exact (true) and complete solution.

The intersection curve computation process in CeDG only requires the use of a generic auxiliary surface and does not need to calculate points related to boundaries or others to obtain the complete solution of the system. The $\mathcal{L}$ function that defines each projection is specific of the modeled system and can be composed of several leaves, as in this example. Each leaf has been represented in a different color in Fig 3.

It is remarkable to note that, in agreement with the small distance between points 1" and 2", almost the entire cylinder lies inside the cone. A slight shift of the cylindrical directrix to the left is enough to convert the bite in a penetration characterized by two unconnected warped curves. Each curve will be defined in such a case for a single leaf. The LSIM – based CeDG model may automatically addresses this change in the nature of the intersection, although the analysis exceeds the scope of this paper.

3.2 Ω defined as interval in $\mathbb{R}$

The second example shows the intersection of a conical surface with a sphere, a situation that can appear in different situations of technical interest. One of them is the illumination of a sphere from a focal light beam, which was defined as case study in Methods section, and it is shown in Fig. 1.

When solving this system, it is necessary to calculate the intersection of the light cone with the illuminating sphere, to obtain a portion of the curve that limits the illuminated area. The curve is completed by a portion of the separatrix circle defined from the light source, which distinguishes the spherical area that can be illuminated

from that which cannot. The spherical illuminated area is therefore defined as the area that can be illuminated (internal to the separatrix circumference) and that which is effectively illuminated by the effective light cone (intersection of light cone - sphere).

The CeDG solution of this system is presented in Fig. 4. The boundary curve of the illuminated surface, given by its projections, is given as the sum of a portion of the separatrix curve (flat) connected by points P_1 and P_2 with the cone - sphere intersection curve (warped).

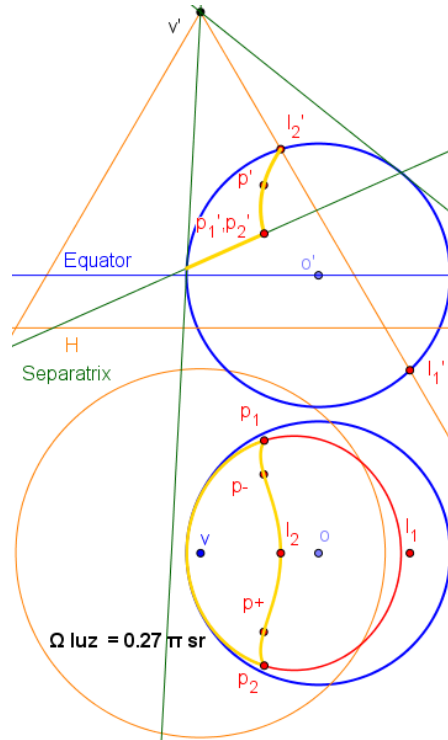


Fig. 4. LSIM – based CeDG solution to the focal illumination of the sphere.

From the complete construction process, we are interested in the LSIM – based solution for the cone – sphere intersection curve. Choosing horizontal planes as auxiliary surfaces, $S(\omega)$, where ω is some real value parameter that allows to identify them, each of them generates a circle in the cone and another in the sphere.

These circles may encounter each other to produce one (tangent circles) or two (secant circles) points of the intersection curve sought. Setting as ω parameter the vertical distance of the horizontal plane in relation to the point I_1 (see Fig. 4), the intersection curve will be given by its projections σ_1 (horizontal) and σ_2 (vertical),

derived from equation (2) for $t = \omega \in \Omega$, being Ω an interval in $\mathbb{R}$ given by $[0, \text{Lim2}]$ with Lim2 equal to the vertical distance of point I_2 with respect to I_1 .

The horizontal projection consists of two leaves:

$$\mathcal{L}_1(t) = \sigma_1 = \begin{cases} \text{locus}(p-, t), & - \text{ semisphere} \\ \text{locus}(p+, t), & + \text{ semisphere} \end{cases} \quad (5)$$

where $p-$ and $p+$ are the horizontal projections of the points obtained by the auxiliary plane.

The vertical projection requires only one leaf as vertical projections $p' += p' -= p'$ for the position used in the model building (Fig. 4):

$$\mathcal{L}_2(t) \equiv \sigma_2 = \text{locus}(p', t) \quad (6)$$

The horizontal plane that produces $P-$ and $P+$ points of the cone – sphere intersection curve is built through a vertical projection (horizontal line) above the point I_1 at a distance equal to the t parameter value, defining t within the interval $[0, \text{Lim2}]$. The value of t is controlled by means of a GeoGebra slider.

Equations (5) and (6) define the cone – sphere intersection curve exactly and completely by means of the functions $\mathcal{L}_1(t)$ and $\mathcal{L}_2(t)$, built on the algebraic entities of the CeDG model. These functions are valid for the position in which the sphere system with focal illumination has been solved (Fig. 1).

To restrict the intersection curve to the portion that is part of the boundary of the illuminated spherical surface, it is enough to replace the lower end (0) of the slider interval t by the vertical distance of p' point (that connects to the separatrix curve) to I'_1 , thus redefining Ω .

As equations (5) and (6) demonstrate, the number of leaves needed to define the projections of the intersecting curve may vary with the direction of projection. A practical consequence of this property is the appearance of an additional leaf for σ_2 by rotating the illuminated sphere and eliminating the symmetry of the system relative to the frontal plane through the center of the sphere. This constructive procedure allows us to generalize the range of the model, and hence of the functions $\mathcal{L}_1(t)$ and $\mathcal{L}_2(t)$, to changes in the projection direction.

This manner, a compound rotation, defined by a horizontal rotation of theta angle, followed by a vertical rotation of phi angle, may be applied to the generator points $P-$ and $P+$, by means of standard descriptive geometry procedures, to transform them first in P_{-h} and P_{+h} , and second in P_{-hv} and P_{+hv} . Applying equation (2) on the final generator points, we obtain the functions $\mathcal{L}_1(t)$ and $\mathcal{L}_2(t)$ that establish σ_1 and σ_2 projections of the intersection curve sphere – cone for a general position controlled by theta and phi rotation angles as follows:

$$\mathcal{L}_1(t) = \sigma_1 = \begin{cases} \text{locus}(p_{-hv}, t), & - \text{ semisphere} \\ \text{locus}(p_{+hv}, t), & + \text{ semisphere,} \end{cases} \quad (7)$$

in which p_{hv}^+ and p_{hv}^- are the horizontal projections of the rotated generator points.

$$\mathcal{L}_2(t) = \sigma_2 = \begin{cases} \text{locus}(p'_{-hv}, t), & - \text{semisphere} \\ \text{locus}(p'_{+hv}, t), & + \text{semisphere}, \end{cases} \quad (8)$$

in which p_{hv}^+ and p_{hv}^- are the vertical projections of the rotated generator points. The CeDG model of the sphere illuminated by the focal beam for the general position given by equations (7-8) will be computed and discussed in the following Section.

4 Results and discussion

The case study has been solved using standard descriptive geometry procedures and the novel LSIM – based CeDG approach, as described in Methods section.

The solution of the curve that defines the boundary of the lightened zone given by standard descriptive geometry techniques is shown in Fig. 5. This is composed by the piece of separatrix circle at the left of P_1 and P_2 points, which connect to the intersection curve between the light cone and sphere. These points are calculated through an auxiliary projection plane parallel to the separatrix plane, where the separatrix circle and the elliptic section that the separatrix plane performs to the cone encounter themselves.

The standard descriptive geometry procedures were executed in the dynamic geometry software (GeoGebra) to avoid differences with respect to the full CeDG approach, except for the cone – sphere intersection, and this manner the ellipse and intersection points are determined as the true objects (algebraically defined).

The cone – sphere intersection warped curve exists between I_1 and I_2 points, which pertain to this one and to the equator of the sphere that is parallel to the vertical plane. We use four additional true points pertaining to the cone – sphere intersection curve, which are P_1 , P_2 , P_3 and P_4 . The last two points have vertical projections equidistant from I_2 and $p'_1 p'_2$. They were obtained through an auxiliary horizontal plane that gives circle sections in sphere and cone, which encounter in P_3 and P_4 . The horizontal and vertical projections of the cone – sphere intersection curve was completed using cubic interpolation splines.

The LSIM – based CeDG solution of the cone – sphere intersection warped curve was presented in Fig. 4. The vertical projection is generated through a locus get by the generator point, p' , and the interval in $\mathbb{R}$ that defines the auxiliary horizontal planes between the vertical dimension of $P_1 = P_2$ and the vertical dimension of I_2 ($y(P_1 = P_2)$, $y(I_2)$). The horizontal projection is given by two symmetrical loci defined through the generator points p^- and p^+ with the same $\mathbb{R}$ interval.

Although the complexity of the LSIM – based CeDG and standard descriptive geometry methods were the same in this first step, the LSIM solution did not require any interpolation curve to be completed.

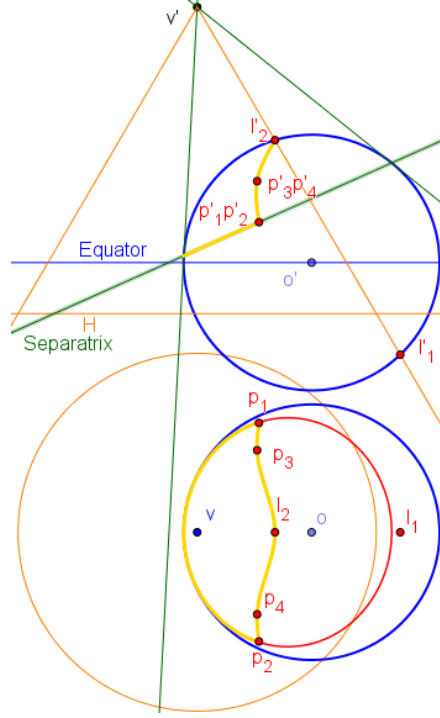


Fig. 5. Interpolation spline – based solution to the focal illumination of the sphere.

Once the system has been computationally modeled, it could be projected in other directions to facilitate the visual inspection and other analysis. This issue cannot be performed repeating the interpolation spline on the new projections of the true intersection points, because, although the interpolation spline could be rotation invariant (if spline's weights are properly selected), the true curvature of the projections' curves depend on the projection direction.

This shortcoming impedes to assure that the standard descriptive geometry solution is compliant with the geometrical requirements of the system, and it compels to build the spline as a function of the projection direction. To the best of our knowledge, the only automatic solution to this deficiency is the use of a CAD approach, but then we lose some of the advantages of descriptive geometry as the underlying basis [18]. The standard descriptive geometry technique has the same problem to guarantee that the solution keeps the compliance with geometrical and physical requirements of the system if the parameters' values vary.

In opposition with standard geometry technique, the change of projection direction may be performed with the LSIM – based CeDG approach, by means of stand-

ard descriptive geometry rotations of the generator points, to give the true new projections of the intersection curve, according to equations (7-8). This is demonstrated in Fig. 6 for the sequence of horizontal (θ) and vertical (ϕ) rotations that are pointed out, where subindex hv refer to the objects after horizontal and vertical rotation.

The locus – based cone – sphere intersection curve projections do not require any interpolation or fitting since they are completely defined through LSIM. The following steps address the accuracy of the LSIM – based solution using the metrics proposed in the Methods Section.

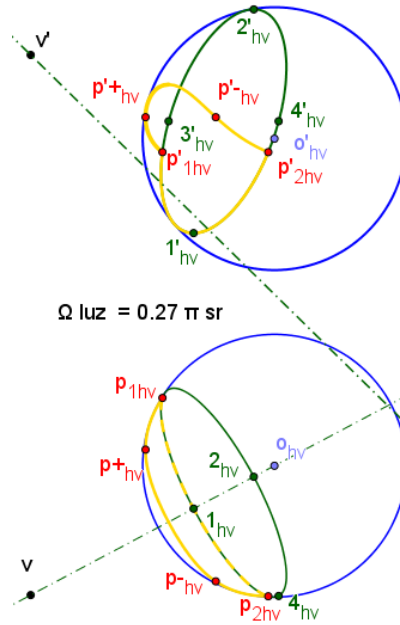


Fig. 6 LSIM – based CeDG solution of focal illumination of the sphere after rotating by $\theta = 49^\circ$ and $\phi = 28^\circ$.

First, we have computed the complete cone – sphere intersection with LSIM and interpolation spline methods.

As we require five true points from the vertical projection of the intersection curve to get the true conic (arc of parabola), and we have currently four true points (I'_2 , $p'_3p'_4$, $p'_1p'_2$ and I'_1), we have added two new true points of the intersection curve, with vertical projections equidistant from I'_2 and $p'_1p'_2$. These points are p'_5 and p'_6 , with the same vertical projection. We have used an auxiliary horizontal plane to get them, in the same manner that we did with $p'_3p'_4$.

After the application of a cubic interpolation spline to obtain the horizontal projection of the cone – sphere intersection for the standard descriptive geometry solution, we lose the symmetry of the system, as shown in Fig. 7.

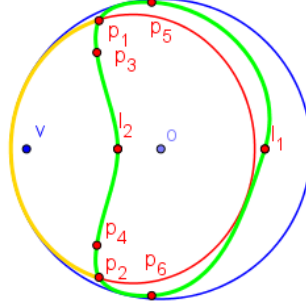


Fig. 7. Interpolation spline – based horizontal projection of the complete cone – sphere intersection.

We need to compel that the tangents to the horizontal interpolation spline in I_1 and I_2 are vertical to keep the required symmetry. The final interpolation spline solution is presented in Fig. 8 (right).

The LSIM – based CeDG based solution to the complete cone – sphere intersection is found simply by the extension of the $\mathbb{R}$ interval to $(y(I_1), y(I_2))$, as shown in Fig. 8 (left).

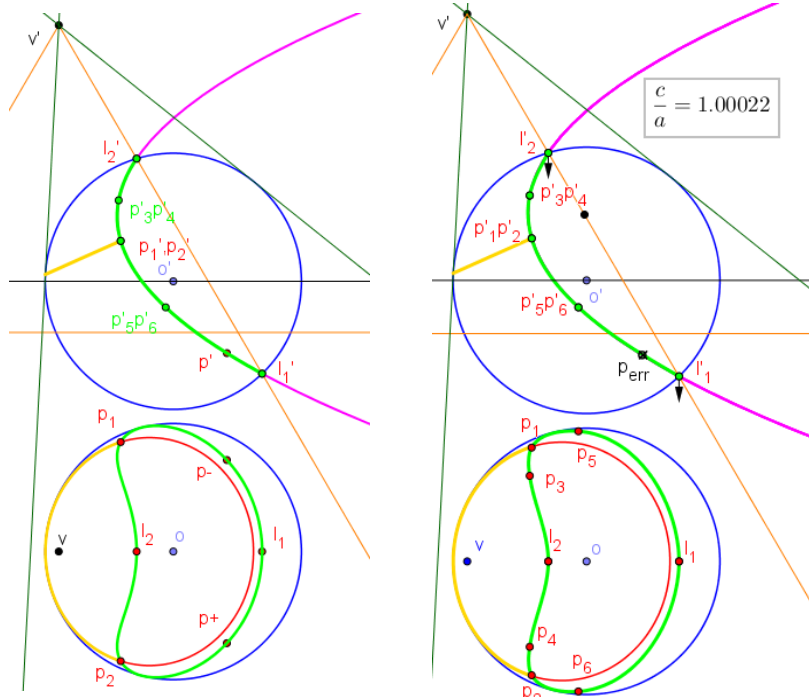


Fig. 8. LSIM – based CeDG (left) and interpolation spline – based (right) solutions of focal illumination of the sphere, where cone – sphere intersection has been completed.

The LSIM – based cone – sphere intersection solution does not require additional conditions. The points (P_3 , P_4 , P_5 , and P_6) were placed on it through the vertical distances (y) in the vertical projection of their homologous points from the interpolation – spline solution of Fig. 8 (right), with the aim of computing the associated conic. As seen, the system solutions in Fig. 8 include also the parabolas given by the vertical projections of the cited five points of cone – sphere intersection in both cases.

The conic eccentricity (c/a , c is the focus – center distance, a is the semi-major axis) was 1.00022 in the interpolation – spline solution versus 1.00028 in the LSIM – based CeDG solution. These values confirm that the CeDG solution keeps the same accuracy than that of the true conic (given from the five true points). The difference among eccentricities' values and the parabola eccentricity (1) is discussed below.

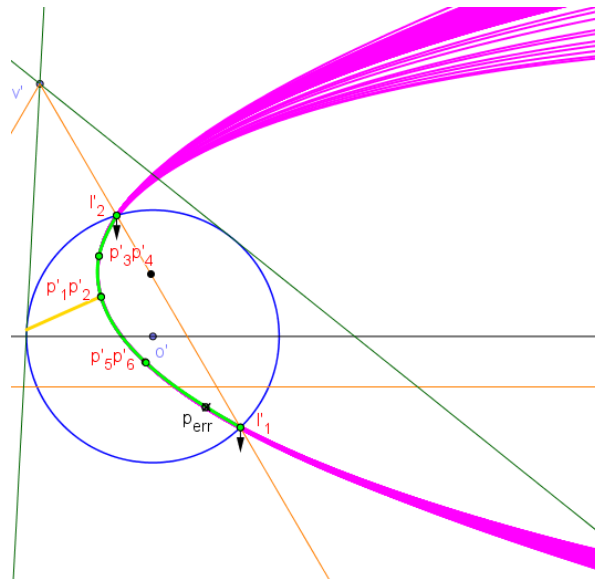


Fig. 9. Conic dispersion in cone – sphere intersection from interpolation spline – based solution.

The dispersion of the conics identified by GeoGebra when their definition points are moved along the cone – sphere intersection projection of the interpolation spline – based solution, as described in Methods, is presented in Fig. 9. This metric is related to the error of the cone – sphere intersection in the standard descriptive geometry technique.

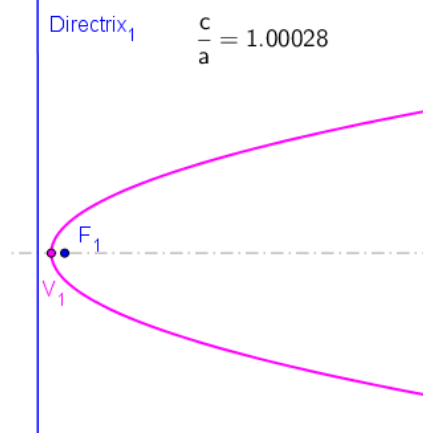


Fig. 10. Conic associated to the cone-sphere intersection obtained by the LSIM – based CeDG approach.

We repeated the same task moving the points of the LSIM – based CeDG solution along the locus defining the vertical projection, but no dispersion was found, as expected.

In agreement with this finding, the true conic is equal to the conic identified through five points from the cone – sphere vertical projection of the CeDG solution, which is presented in Fig. 10. This is the reference conic that defines the arc of parabola of the cone – sphere vertical projection. We include its directrix, focus and vertex.

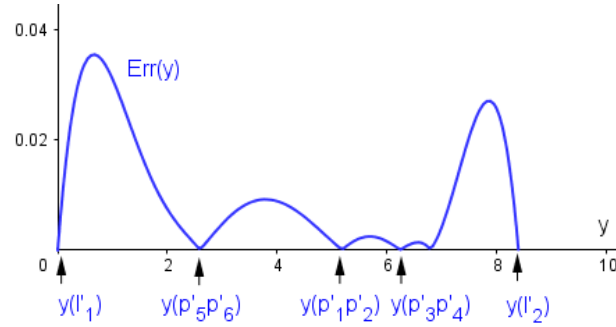


Fig. 11. Horizontal absolute error (cm) of the vertical projection of the interpolation spline – based cone – sphere solution, as a function of the vertical dimension.

The horizontal distance between the vertical projection of interpolation spline – based cone – sphere solution and the true conic, as a function of the vertical distance, $Err(y)$, was computed for the system model presented in Fig. 8 (right, by means of points marked with P_{err}) and it is shown in Fig. 11.

As expected, the errors of the true five points pertaining to the vertical projection of the cone – sphere intersection are null. Their vertical distances are marked in the abscissa axis of Fig. 11. The maximum absolute error is 0.35 mm (0.71 % with respect to the horizontal distance between I'_1 and I'_2).

Although the errors are not large, in agreement with projections presented in Fig. 8 (right), they could increase in other projections. Previous shortcomings in the geometrical and physical compliance, together with this dimensional error, reduce the capability of the standard descriptive geometry technique to extend the solution to other parameters' values and projections, in opposition to LSIM – based CeDG.

We have finally evaluated the influence of the model's parameters on the $Err(y)$ dimensional error of the interpolation spline – based cone – sphere solution for the same projection. The Fig. 12 shows how the light beam focus radius, defined by $rBCono$ parameter, affects to $Err(y)$ when it goes down from 70 mm to 60 mm (red trajectories), and when it goes up from 70 mm to 80 mm (blue trajectories). The maximum absolute errors were 0.32 mm (0.72 %) for $rBCono = 60$ mm and 0.38 mm (0.74 %) for $rBCono = 80$ mm. The values of the vertical distances marked in the abscissa axis refer to $rBCono = 70$ mm.

The dimensional error behavior is similar for the other parameters. Therefore, although the interpolation spline – based cone – sphere intersection solution has limitations that difficult the extension to other parameters' values and projections when the compliance with geometrical and physical requirements is kept, the model is robust against perturbations in these parameters.

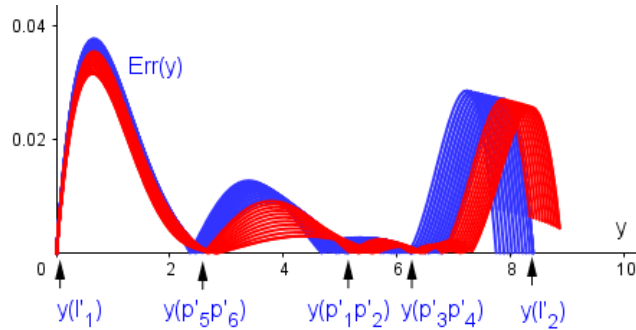


Fig. 12. Influence of the light beam focus radius on the horizontal absolute error of the interpolation spline – based cone – sphere intersection (60 – 70 mm red; 70 – 80 mm blue, increments of 1mm).

The difference between the conic eccentricity (c/a) of the reference parabola (1.00028) and the true parabola (1) is associated to the accuracy of GeoGebra to solve the 5 – equations system defined by the 5 points that define the parabola coefficients. The relative error is 0.028 %. Although it is a small error, it affects to the

capability of CeDG to control the integrity of geometry ($c/a > 1$ is a hyperbola) and the accuracy and thus more research is required to improve this issue.

5 Conclusions

CeDG is a novel approach to computer modelling of 3D geometric systems based on descriptive geometry, which overcomes several limitations of current CAD systems. This study has presented first the theory underlying the Locus – based Surfaces' Intersection Method (LSIM), and second a case study about the focal illumination of a sphere that allowed comparing LSIM against standard descriptive geometry procedures.

The outcomes demonstrate that the complexity of the LSIM CeDG solution, in terms of the number of iterations required, is lesser than that of the standard descriptive geometry procedures. In addition, the accuracy of LSIM CeDG was much higher than that of standard procedures. In fact, the accuracy was only limited by the numerical precision of GeoGebra, since the computed intersection curve is formulated through algebraic objects that keep the geometric integrity of the curve projections. We may conclude that CeDG provides the true cone – sphere intersection curve.

Finally, the LSIM CeDG solution kept the compliance with geometrical requirements when the projection and parameters' values of the model were changed, in opposition with the standard technique that requires additional geometrical restrictions.

CeDG is currently in a starting version that will evolve to include the automation and improvement of many geometric procedures to facilitate its diffusion to professional and academic domains.

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